Hybrid Cloud with Provider-Managed Components

NetApp Solutions

NetApp
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NetApp Hybrid Multicloud solutions for Red Hat OpenShift Container workloads

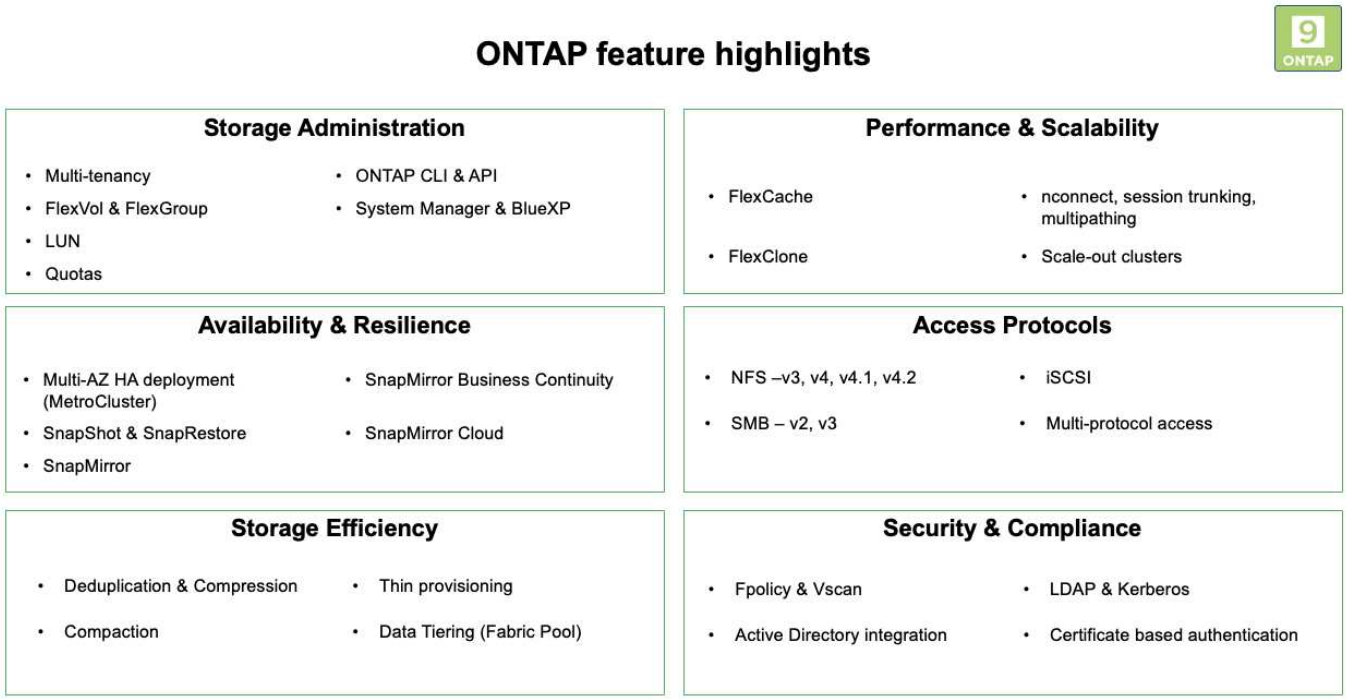
Overview

NetApp is seeing a significant increase in customers modernizing their legacy enterprise applications and building new applications using containers and orchestration platforms built around Kubernetes. Red Hat OpenShift Container Platform is one example that we see adopted by many of our customers.

As more and more customers begin adopting containers within their enterprises, NetApp is perfectly positioned to help serve the persistent storage needs of their stateful applications and classic data management needs such as data protection, data security, and data migration. However, these needs are met using different strategies, tools, and methods.

NetApp ONTAP based storage options listed below, deliver security, data protection, reliability, and flexibility for containers and Kubernetes deployments.

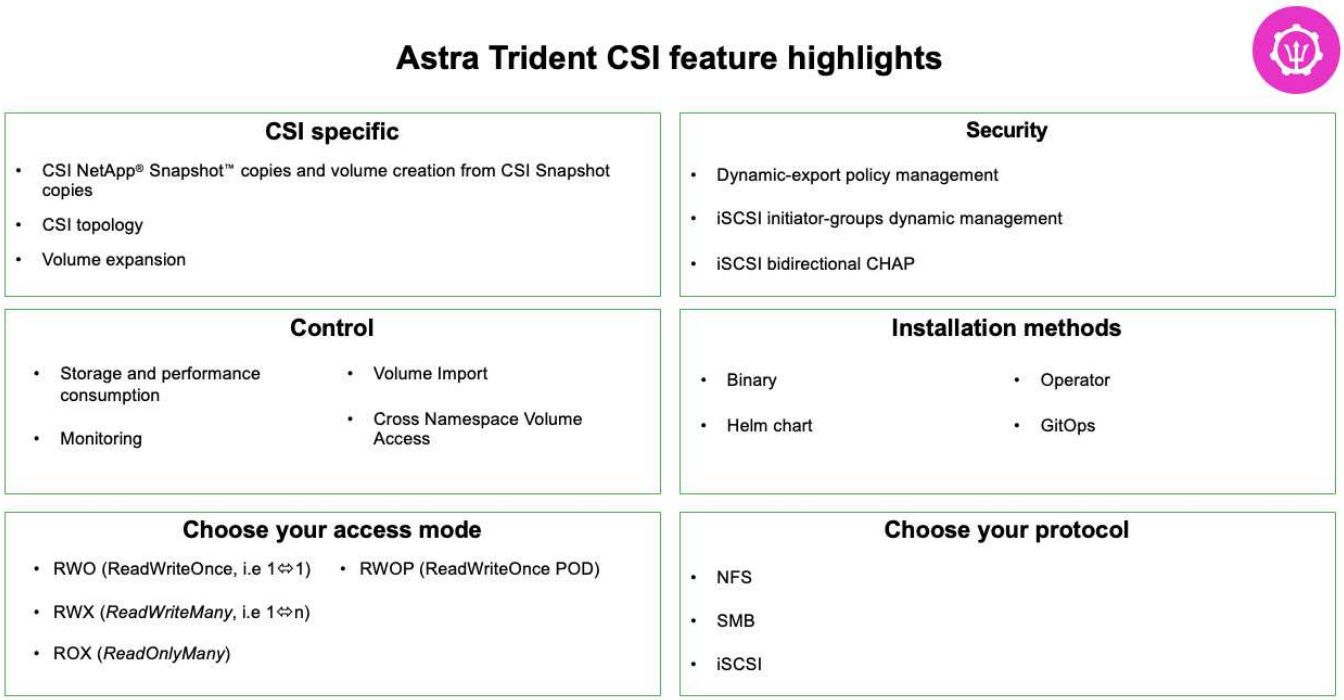
- Self-managed storage in on-premises:
  - NetApp Fabric Attached Storage (FAS), NetApp All Flash FAS Arrays (AFF), NetApp All SAN Array (ASA) and ONTAP Select
- Provider-managed storage in on-premises:
  - NetApp Keystone provides Storage as a Service (STaaS)
- Self-managed storage in the cloud:
  - NetApp Cloud Volumes ONTAP(CVO) provide self managed storage in the hyperscalers
- Provider-managed storage in the cloud:
  - Cloud Volumes Service for Google Cloud (CVS), Azure NetApp Files (ANF), Amazon FSx for NetApp ONTAP offer fully managed storage in the hyperscalers



NetApp BlueXP enables you to manage all of your storage and data assets from a single control plane/interface.

You can use BlueXP to create and administer cloud storage (for example, Cloud Volumes ONTAP and Azure NetApp Files), to move, protect, and analyze data, and to control many on-prem and edge storage devices.

NetApp Astra Trident is a CSI Compliant Storage Orchestrator that enable quick and easy consumption of persistent storage backed by a variety of the above-mentioned NetApp storage options. It is an open-source software maintained and supported by NetApp.



Business critical container workloads need more than just persistent volumes. Their data management requirements require protection and migration of the application kubernetes objects as well.

> Application data includes kubernetes objects in addition to the user data: Some examples are as follows:
> - kubernetes objects such as pods specs, PVCs, deployments, services
> - custom config objects such as config maps and secrets
> - persistent data such as Snapshot copies, backups, clones
> - custom resources such as CRs and CRDs

NetApp Astra Control, available as both fully-managed and self-managed software, provides orchestration for robust application data management. Refer to the Astra documentation for additional details on the Astra family of products.

This reference documentation provides validation of migration and protection of container-based applications, deployed on RedHat OpenShift container platform, using NetApp Astra Control Center. In addition, the solution provides high-level details for the deployment and the use of Red Hat Advanced Cluster Management (ACM) for managing the container platforms. The document also highlights the details for the integration of NetApp storage with Red Hat OpenShift container platforms using Astra Trident CSI provisioner. Astra Control Center is deployed on the hub cluster and is used to manage the container applications and their persistent storage lifecycle. Finally, it provides a solution for replication and failover and fail-back for container workloads on managed Red Hat OpenShift clusters in AWS (ROSA) using Amazon FSx for NetApp ONTAP (FSxN) as

persistent storage.

NetApp Solution with Managed Red Hat OpenShift Container platform workloads on AWS

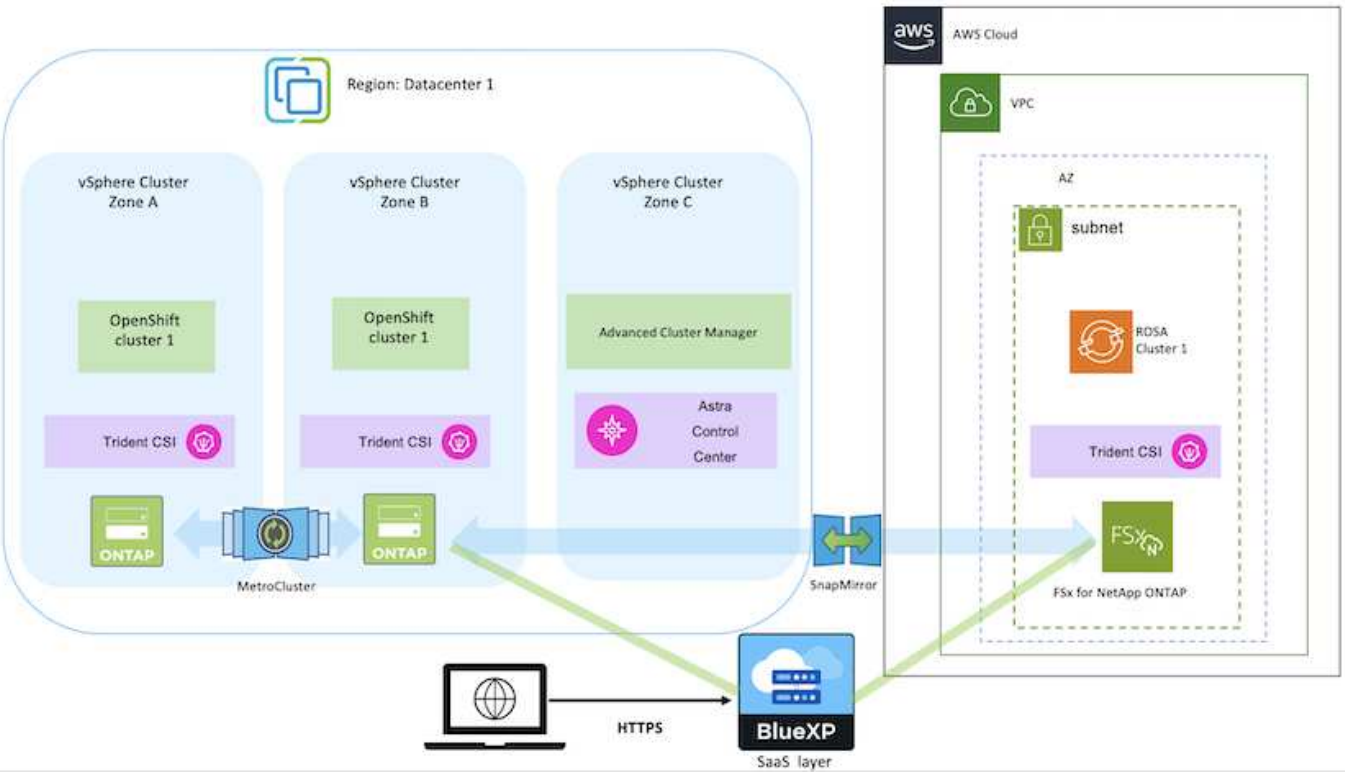
Customers may be "born in the cloud" or may be at a point in their modernization journey when they are ready to move some select workloads or all workloads from their data centers to the cloud. They may choose to use provider-managed OpenShift containers and provider-managed NetApp storage in the cloud for running their workloads. They should plan and deploy the Managed Red Hat OpenShift container clusters (ROSA) in the cloud for a successful production-ready environment for their container workloads. When they are in AWS cloud, they could also deploy FSx for NetApp ONTAP for the storage needs.

FSx for NetApp ONTAP delivers data protection, reliability, and flexibility for container deployments in AWS. Astra Trident serves as the dynamic storage provisioner to consume the persistent FSxN storage for customers' stateful applications.

As ROSA can be deployed in HA mode with control plane nodes spread across multiple availability zones, FSx ONTAP can also be provisioned with Multi-AZ option which provides high availability and protect against AZ failures.

> There are no data transfer charges when accessing an Amazon FSx file system from the file system's preferred Availability Zone (AZ). For more info on pricing, refer here.

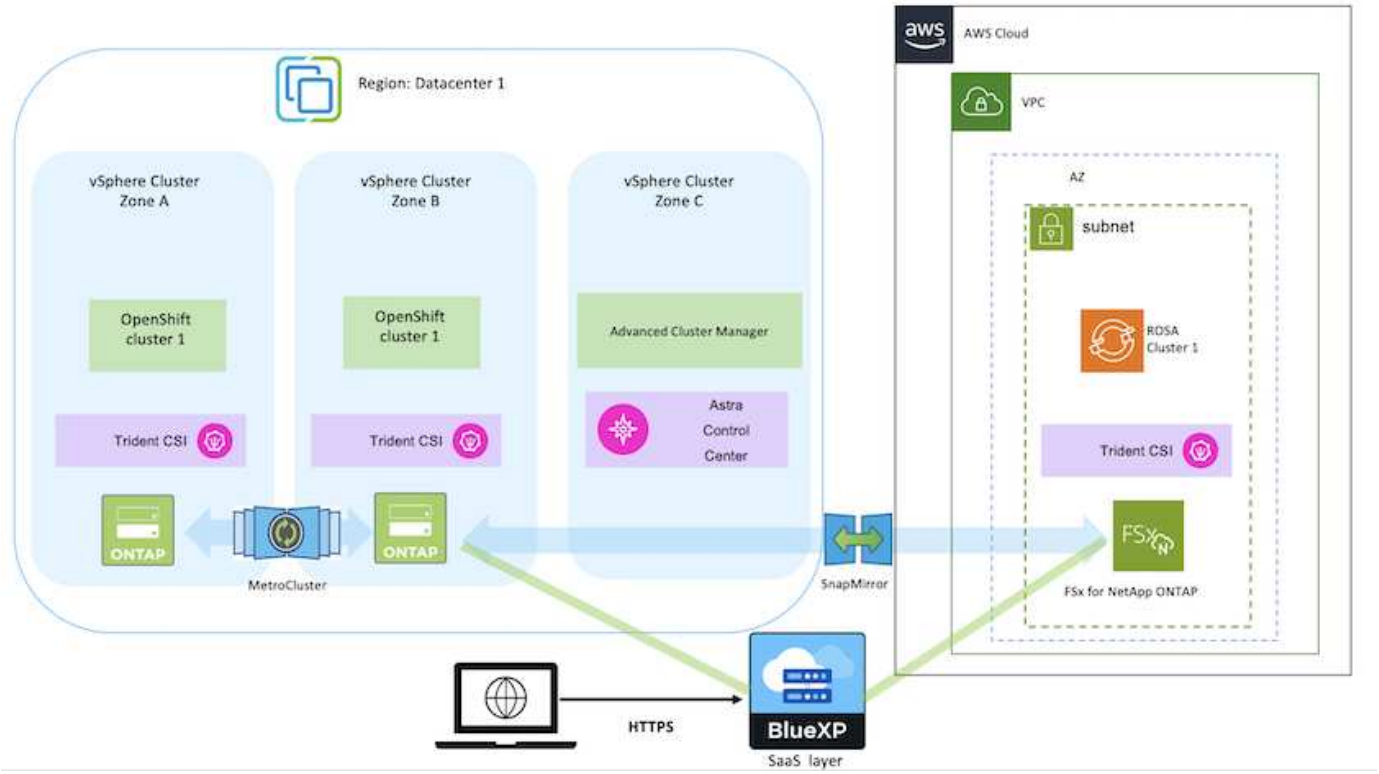
Data protection and migration solution for OpenShift Container workloads



Deploy and configure the Managed Red Hat OpenShift Container platform on AWS

This section describes a high-level workflow of setting up the Managed Red Hat OpenShift clusters on AWS(ROSA). It shows the use of Managed FSx for NetApp ONTAP (FSxN) as the storage backend by Astra Trident to provide persistent volumes. Details are provided about the deployment of FSxN on AWS using BlueXP. Also, details are provided about the use of BlueXP and OpenShift GitOps (Argo CD) to perform data protection and migration activities for the stateful applications on ROSA clusters.

Here is a diagram that depicts the ROSA clusters deployed on AWS and using FSxN as the backend storage.



This solution was verified by using two ROSA clusters in two VPCs in AWS. Each ROSA cluster was integrated with FSxN using Astra Trident. There are several ways of deploying ROSA clusters and FSxN in AWS. This high-level description of the setup provides documentation links for the specific method that was used. You can refer to the other methods in the relevant links provided in the resources section.

The setup process can be broken down into the following steps:

Install ROSA clusters

- Create two VPCs and set up VPC peering connectivity between the VPCs.
- Refer here for instructions to install ROSA clusters.

Install FSxN

- Install FSxN on the VPCs from BlueXP.
  Refer here for BlueXP account creation and to get started.
  Refer here for installing FSxN.
  Refer here for creating a connector in AWS to manage the FSxN.
- Deploy FSxN using AWS.
  Refer here for deployment using AWS console.

Install Trident on ROSA clusters (using Helm chart)

- Use Helm chart to install Trident on ROSA clusters.
  url for the Helm chart: https://netapp.github.io/trident-helm-chart

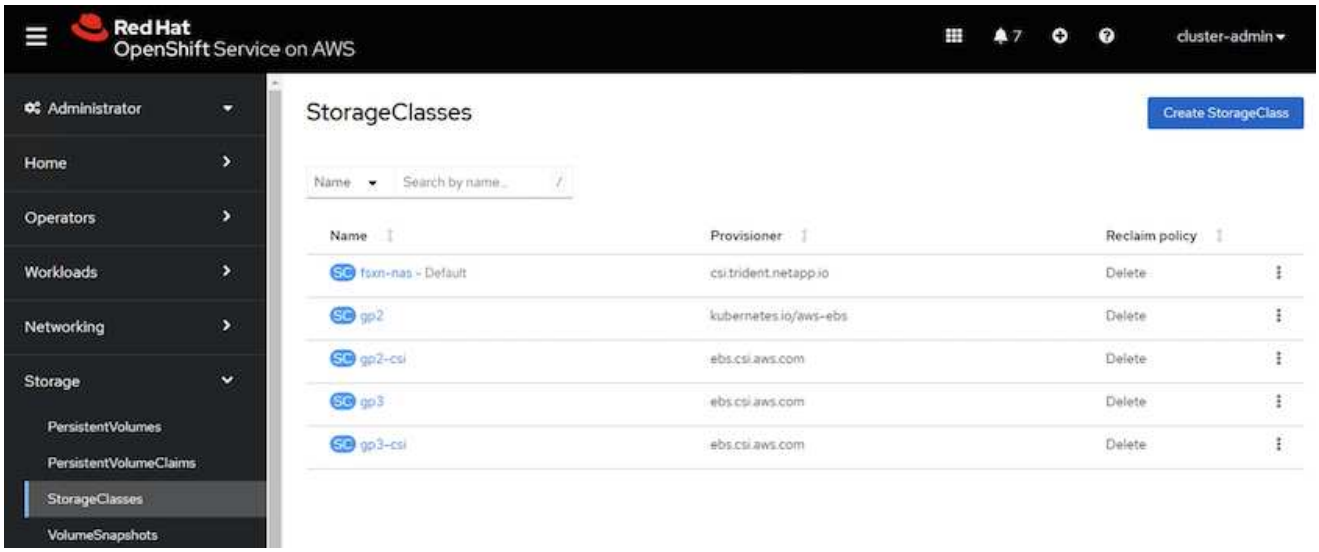
Integration of FSxN with Astra Trident for ROSA clusters

> OpenShift GitOps can be utilized to deploy Astra Trident CSI to all managed clusters as they get registered to ArgoCD using ApplicationSet.

```yaml
apiVersion: argoproj.io/v1alpha1
kind: ApplicationSet
metadata:
  name: trident-operator
spec:
  generators:
  - clusters: {}
    # selector:
    #   matchLabels:
    #     tridentversion: '23.04.0'
  template:
    metadata:
      name: '{{nameNormalized}}-trident'
    spec:
      destination:
        namespace: trident
        server: '{{server}}'
      source:
        repoURL: 'https://netapp.github.io/trident-helm-chart'
        targetRevision: 23.04.0
        chart: trident-operator
      project: default
      syncPolicy:
        syncOptions:
          - CreateNamespace=true
```

Create backend and storage classes using Trident (for FSxN)

- Refer here for details about creating backend and storage class.
- Make the storage class created for FsxN with Trident CSI as default from OpenShift Console. See screenshot below:



Deploy an application using OpenShift GitOps (Argo CD)

- Install OpenShift GitOps operator on the cluster. Refer to instructions here.
- SetUp a new Argo CD instance for the cluster. Refer to instructions here.

Open the console of Argo CD and deploy an app.
As an example, you can deploy a Jenkins App using Argo CD with a Helm Chart.
When creating the application, the following details were provided:
Project: default
cluster: https://kubernetes.default.svc
Namespace: Jenkins
The url for the Helm Chart: https://charts.bitnami.com/bitnami

Helm Parameters:
global.storageClass: fsxn-nas

Data protection

This page shows the data protection options for Managed Red Hat OpenShift on AWS (ROSA) clusters using Astra Control Service. Astra Control Service (ACS) provides an easy-to-use graphical user-interface with which you can add clusters, define applications running on them, and perform application aware data management activities. ACS functions can also be accessed using an API that allows for automation of workflows.

Powering Astra Control (ACS or ACC) is NetApp Astra Trident. Astra Trident integrates several types of Kubernetes clusters such as Red Hat OpenShift, EKS, AKS, SUSE Rancher, Anthos etc., with various flavors

of NetApp ONTAP storage such as FAS/AFF, ONTAP Select, CVO, Google Cloud Volumes Service, Azure NetApp Files and Amazon FSx for NetApp ONTAP.

This section provides details for the following data protection options using ACS:

- A video showing Backup and Restore of a ROSA application running in one region and restoring to another region.
- A video showing Snapshot and Restore of a ROSA application.
- Step-by-step details of installing a ROSA cluster, Amazon FSx for NetApp ONTAP, using NetApp Astra Trident to integrate with storage backend, installing a postgresql application on ROSA cluster, using ACS to create a snapshot of the application and restoring the application from it.
- A blog showing step-by-step details of creating and restoring from a snapshot for a mysql application on a ROSA cluster with FSx for ONTAP using ACS.

Backup/Restore from Backup

The following video shows the backup of a ROSA application running in one region and restoring to another region.

FSx NetApp ONTAP for Red Hat OpenShift Service on AWS

Snapshot/Restore from snapshot

The following video shows taking a snapshot of a ROSA application and restoring from the snapshot after.

Snapshot/Restore for Applications on Red Hat OpenShift Service on AWS (ROSA)clusters with Amazon FSx for NetApp ONTAP storage

Blog

- Using Astra Control Service for data management of apps on ROSA clusters with Amazon FSx storage

Step-by-Step Details to create snapshot and restore from it

Prerequisite setup

- AWS account
- Red Hat OpenShift account
- IAM user with appropriate permissions to create and access ROSA cluster
- AWS CLI
- ROSA CLI
- OpenShift CLI(oc)
- VPC with subnets and appropriate gateways and routes
- ROSA Cluster installed into the VPC
- Amazon FSx for NetApp ONTAP created in the same VPC
- Access to the ROSA cluster from OpenShift Hybrid Cloud Console

Next Steps

1. Create an admin user and login to the cluster.
2. Create a kubeconfig file for the cluster.
3. Install Astra Trident on the cluster.
4. Create a backend, storage class and snapshot class configuration using the Trident CSI provisioner.
5. Deploy a postgresql application on the cluster.
6. Create a database and add a record.
7. Add the cluster into ACS.
8. Define the application in ACS.
9. Create a snapshot using ACS.
10. Delete the database in the postgresql application.
11. Restore from a snapshot using ACS.
12. Verify your app has been restored form the snapshot.

1. Create an admin user and login to the cluster

Access the ROSA cluster by creating an admin user with the following command : (You need to create an admin user only if you did not create one at the time of installation)

```
rosa create admin --cluster=<cluster-name>
```

The command will provide an output that will look like the following. Login to the cluster using the `oc login` command provided in the output.

```
W: It is recommended to add an identity provider to login to this cluster.
See 'rosa create idp --help' for more information.
I: Admin account has been added to cluster 'my-rosa-cluster'. It may take up
to a minute for the account to become active.
I: To login, run the following command:
oc login https://api.my-rosa-cluster.abcd.p1.openshiftapps.com:6443 \
--username cluster-admin \
--password FWGYL-2mkJI-00000-00000
```

> You can also login to the cluster using a token. If you already created an admin-user at the time of cluster creation, you can login to the cluster from the Red Hat OpenShift Hybrid Cloud console with the admin-user credentials. Then by clicking on the top right corner where it displays the name of the logged in user, you can obtain the `oc login` command (token login) for the command line.

2. Create a kubeconfig file for the cluster

Follow the procedures here to create a kubeconfig file for the ROSA cluster. This kubeconfig file will be used later when you add the cluster into ACS.

3. Install Astra Trident on the cluster

Install Astra Trident (latest version) on the ROSA cluster. To do this, you can follow any one of the procedures given here. To install Trident using helm from the console of the cluster, first create a project called Trident.

Then from the Developer view, create a Helm chart repository. For the URL field use `'https://netapp.github.io/trident-helm-chart'`. Then create a helm release for Trident operator.

Project: trident

Create Helm Chart Repository

Add helm chart repository.

Configure via: (●) Form view ( ) YAML view

Scope type

( ) Namespaced scoped (ProjectHelmChartRepository)

Add Helm Chart Repository in the selected namespace.

(●) Cluster scoped (HelmChartRepository)

Add Helm Chart Repository at the cluster level and in all namespaces.

Name *

trident

A unique name for the Helm Chart repository.

Display name

Astra Trident

A display name for the Helm Chart repository.

Description

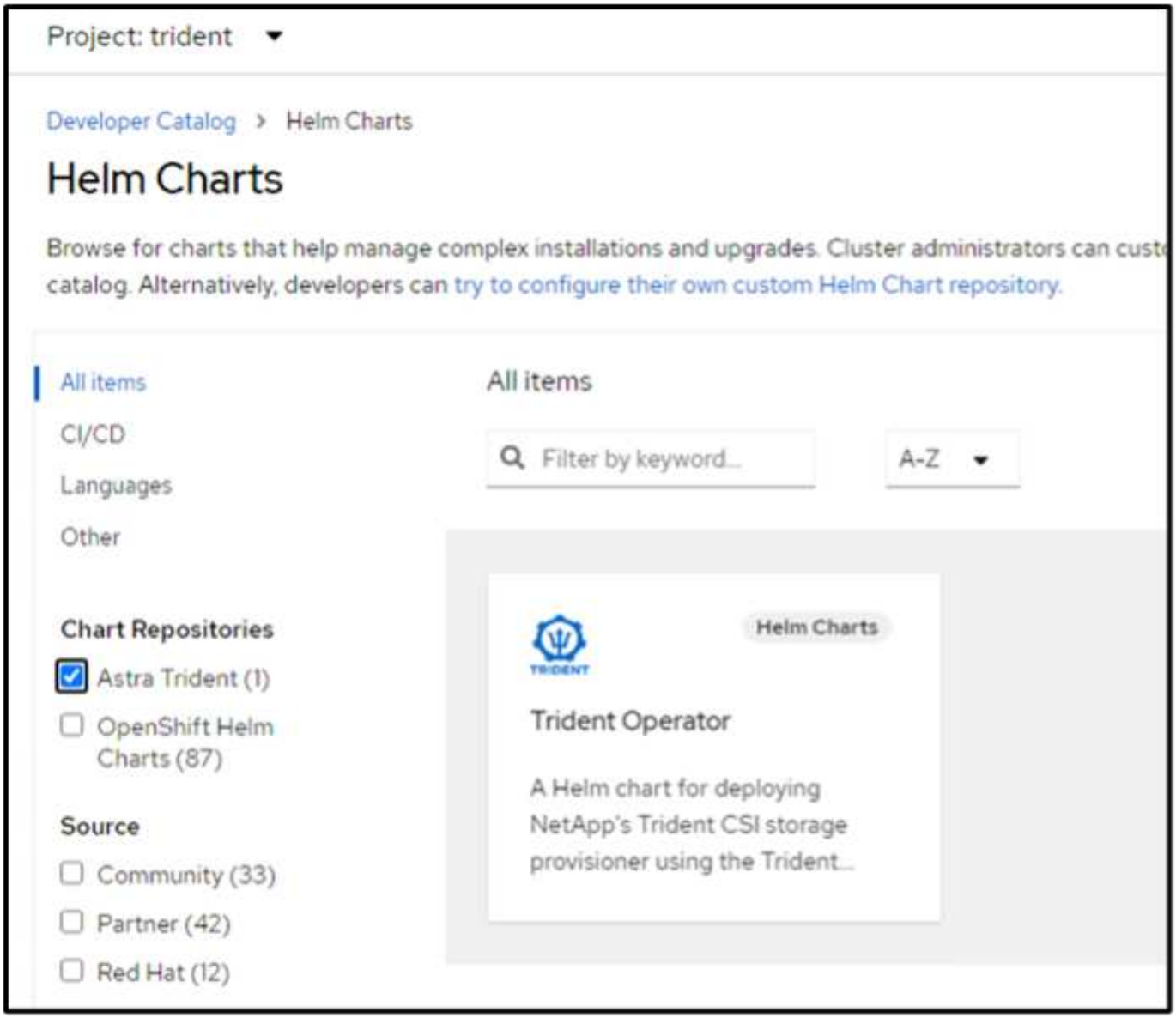
NetApp Astra Trident

A description for the Helm Chart repository.

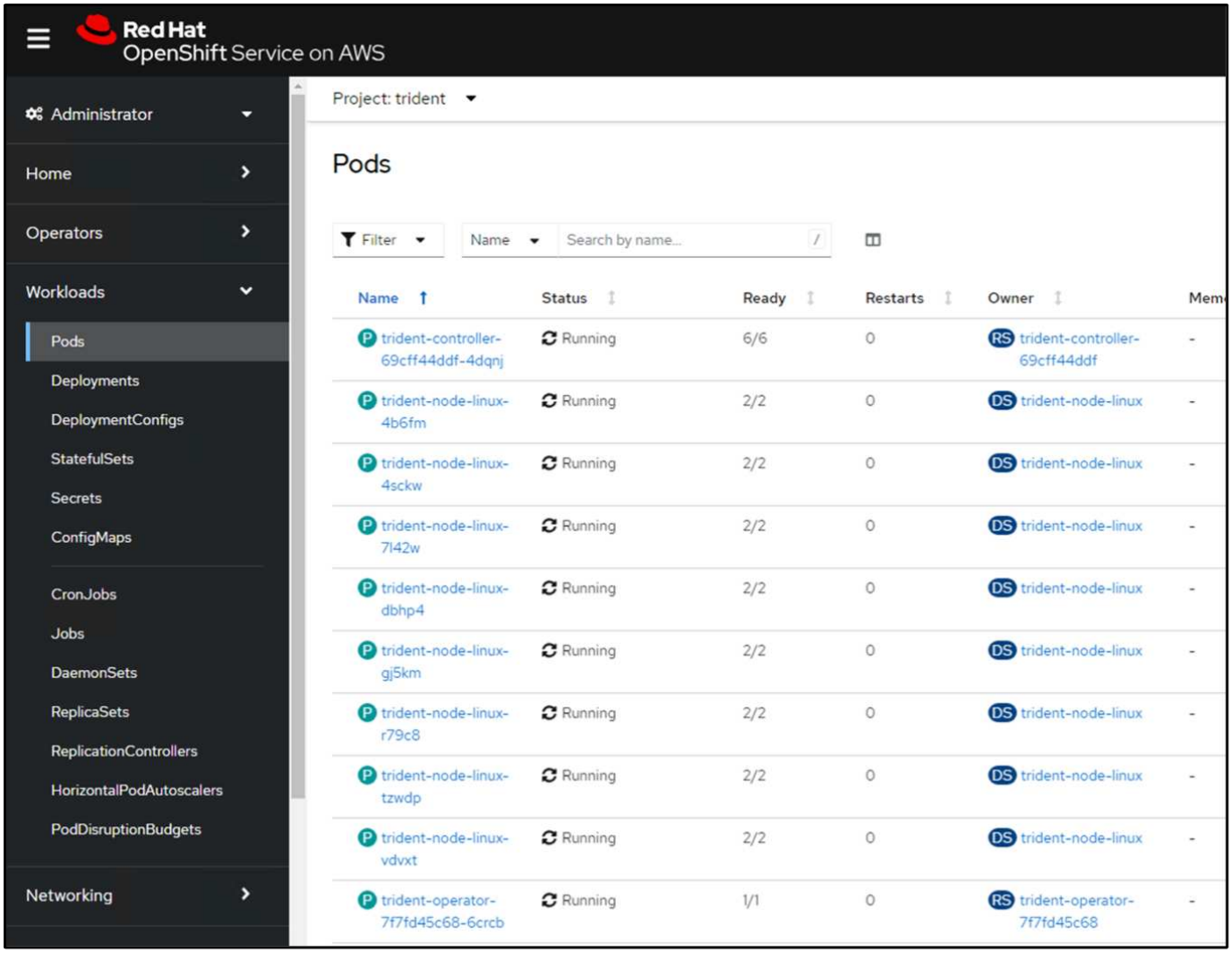
[ ] Disable usage of the repo in the developer catalog.

URL *

https://netapp.github.io/trident-helm-chart



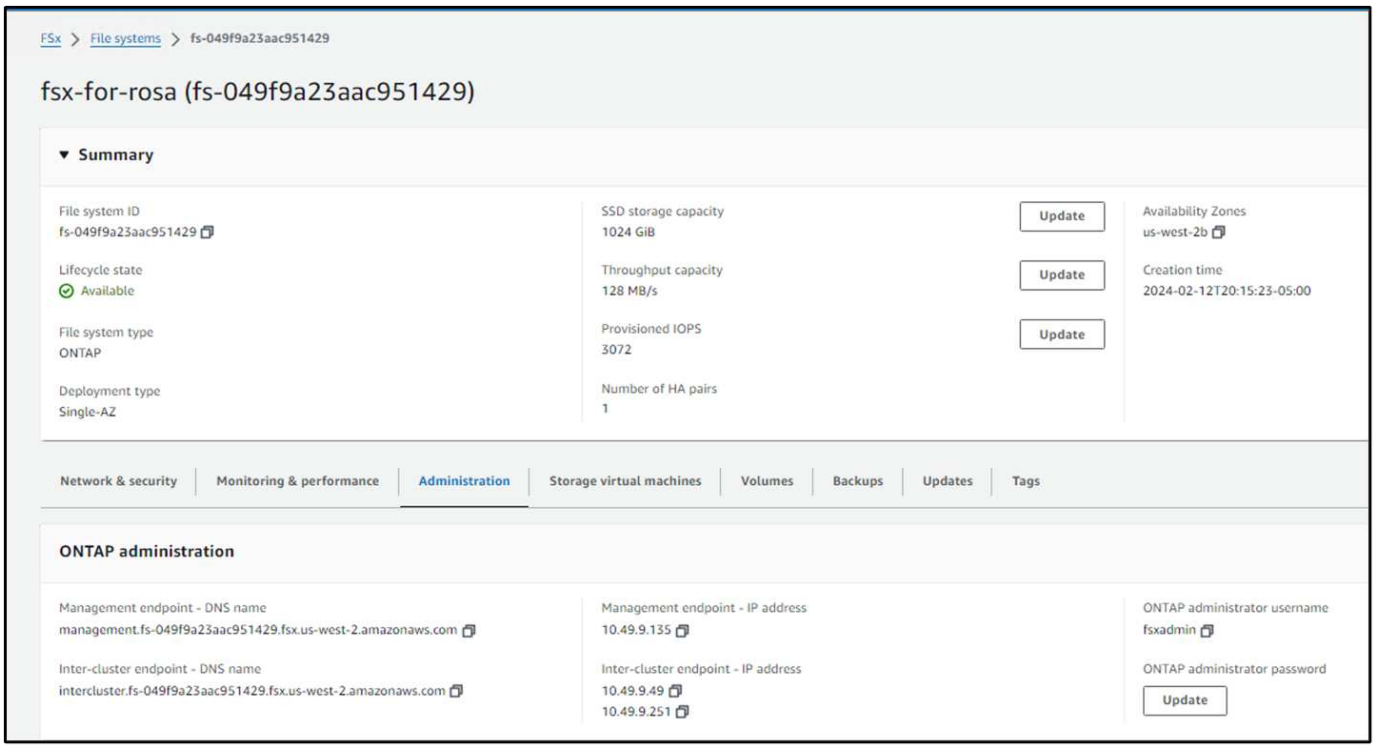
Verify all trident pods are running by going back to the Administrator view on the console and selecting pods in the trident project.



4. Create a backend, storage class and snapshot class configuration using the Trident CSI provisioner

Use the yaml files shown below to create a trident backend object, storage class object and the Volumesnapshot object. Be sure to provide the credentials to your Amazon FSx for NetApp ONTAP file system you created, the management LIF and the vserver name of your file system in the configuration yaml for the backend. To get those details, go to the AWS console for Amazon FSx and select the file system, navigate to the Administration tab. Also, click on update to set the password for the `fsxadmin` user.

> You can use the command line to create the objects or create them with the yaml files from the hybrid cloud console.



Trident Backend Configuration

```yaml
apiVersion: v1
kind: Secret
metadata:
  name: backend-tbc-ontap-nas-secret
type: Opaque
stringData:
  username: fsxadmin
  password: <password>
---
apiVersion: trident.netapp.io/v1
kind: TridentBackendConfig
metadata:
  name: ontap-nas
spec:
  version: 1
  storageDriverName: ontap-nas
  managementLIF: <management lif>
  backendName: ontap-nas
  svm: fsx
  credentials:
    name: backend-tbc-ontap-nas-secret
```

Storage Class

```yaml
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ontap-nas
provisioner: csi.trident.netapp.io
parameters:
  backendType: "ontap-nas"
  media: "ssd"
  provisioningType: "thin"
  snapshots: "true"
allowVolumeExpansion: true
```

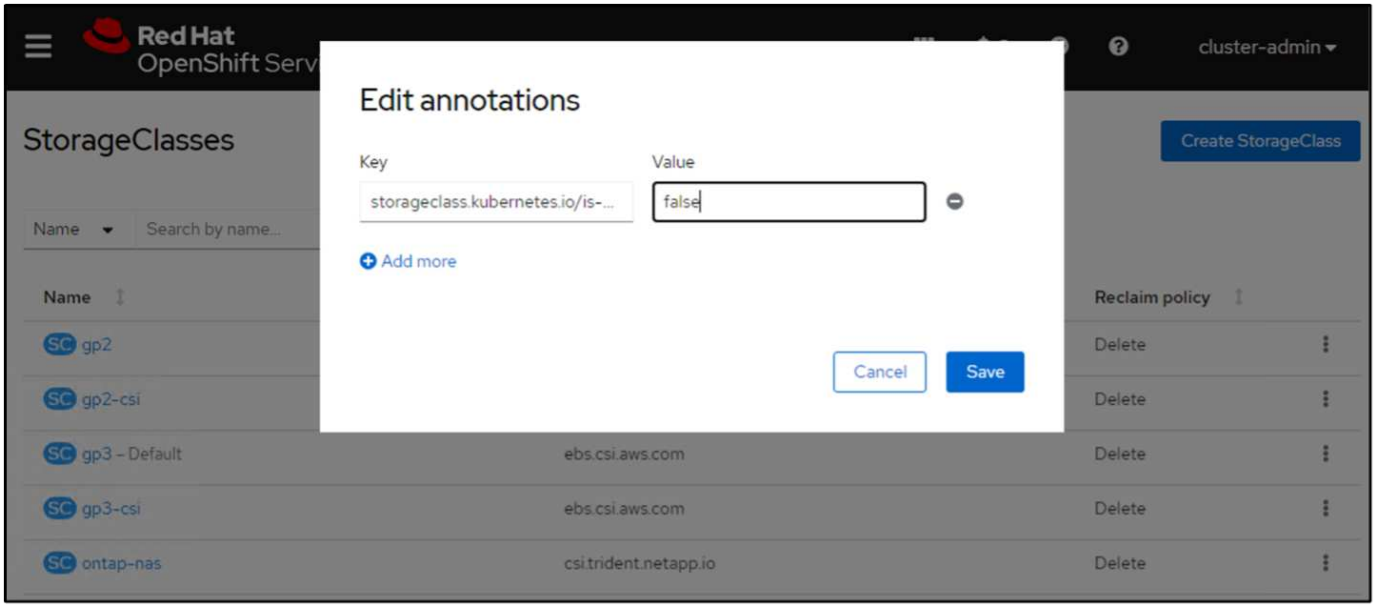
snapshot class

```yaml
apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshotClass
metadata:
  name: trident-snapshotclass
driver: csi.trident.netapp.io
deletionPolicy: Delete
```

Verify that the backend, storage class and the trident-snapshotclass objects are created by issuing the commands shown below.

```
[ec2-user@ip-10-49-11-132 storage]$ kubectl get tbc -n trident
NAME        BACKEND NAME   BACKEND UUID                           PHASE   STATUS
ontap-nas   ontap-nas      8a5e4583-2dac-46bb-b01e-fa7c3816f121   Bound   Success
[ec2-user@ip-10-49-11-132 storage]$ kubectl get sc
NAME            PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
gp2             kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer   true                   3h23m
gp2-csi         ebs.csi.aws.com         Delete          WaitForFirstConsumer   true                   3h19m
gp3 (default)   ebs.csi.aws.com         Delete          WaitForFirstConsumer   true                   3h23m
gp3-csi         ebs.csi.aws.com         Delete          WaitForFirstConsumer   true                   3h19m
ontap-nas       csi.trident.netapp.io   Delete          Immediate              true                   141m
[ec2-user@ip-10-49-11-132 storage]$ kubectl get Volumesnapshotclass
NAME                    DRIVER                  DELETIONPOLICY   AGE
csi-aws-vsc             ebs.csi.aws.com         Delete           3h19m
trident-snapshotclass   csi.trident.netapp.io   Delete           6m56s
[ec2-user@ip-10-49-11-132 storage]$
```

At this time, an important modification you need to make is to set ontap-nas as the default storage class instead of gp3 so that the postgresql app you deploy later can use the default storage class. In the Openshift console of your cluster, under Storage select StorageClasses. Edit the annotation of the current default class to be false and add the annotation storageclass.kubernetes.io/is-default-class set to true for the ontap-nas storage class.



5. Deploy a postgresql application on the cluster

You can deploy the application from the command line as follows:

```
helm install postgresql bitnami/postgresql -n postgresql --create-namespace
```

```
[ec2-user@ip-10-49-11-132 astra]$ helm install postgresql bitnami/postgresql -n postgresql --create-namespace
NAME: postgresql
LAST DEPLOYED: Tue Feb 13 14:46:16 2024
NAMESPACE: postgresql
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
CHART NAME: postgresql
CHART VERSION: 14.0.4
APP VERSION: 16.2.0

** Please be patient while the chart is being deployed **

PostgreSQL can be accessed via port 5432 on the following DNS names from within your cluster:

    postgresql.postgresql.svc.cluster.local - Read/Write connection

To get the password for "postgres" run:

    export POSTGRES_PASSWORD=$(kubectl get secret --namespace postgresql postgresql -o jsonpath="{.data.postgres-password}" | base64 -d)

To connect to your database run the following command:

    kubectl run postgresql-client --rm --tty -i --restart='Never' --namespace postgresql --image docker.io/bitnami/postgresql:16.2.0-debian-11-r1 --env="PG
PASSWORD=$POSTGRES_PASSWORD" \
      --command -- psql --host postgresql -U postgres -d postgres -p 5432

    > NOTE: If you access the container using bash, make sure that you execute "/opt/bitnami/scripts/postgresql/entrypoint.sh /bin/bash" in order to avoid
the error "psql: local user with ID 1001} does not exist"

To connect to your database from outside the cluster execute the following commands:

    kubectl port-forward --namespace postgresql svc/postgresql 5432:5432 &
    PGPASSWORD="$POSTGRES_PASSWORD" psql --host 127.0.0.1 -U postgres -d postgres -p 5432

WARNING: The configured password will be ignored on new installation in case when previous PostgreSQL release was deleted through the helm command. In that
 case, old PVC will have an old password, and setting it through helm won't take effect. Deleting persistent volumes (PVs) will solve the issue.
[ec2-user@ip-10-49-11-132 astra]$
```

If you do not see the application pods running, then there might be an error caused due to security context constraints.

```
[ec2-user@ip-10-49-11-132 astra]$ kubectl get all -n postgresql
NAME                    TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
service/postgresql      ClusterIP   172.30.245.50   <none>        5432/TCP   12m
service/postgresql-hl   ClusterIP   None            <none>        5432/TCP   12m

NAME                          READY   AGE
statefulset.apps/postgresql   0/1     12m
[ec2-user@ip-10-49-11-132 astra]$ kubectl get events -n postgresql
LAST SEEN   TYPE      REASON                  OBJECT                                  MESSAGE
2m39s       Normal    WaitForFirstConsumer    persistentvolumeclaim/data-postgresql-0   waiting for first consumer to be created before binding
12m         Normal    SuccessfulCreate        statefulset/postgresql                  create Claim data-postgresql-0 Pod postgresql-0 in StatefulSet postg
resql success
107s        Warning   FailedCreate            statefulset/postgresql                  create Pod postgresql-0 in StatefulSet postgresql failed error: pods
 "postgresql-0" is forbidden: unable to validate against any security context constraint: [provider "trident-controller": Forbidden: not usable by user or
serviceaccount, provider "anyuid": Forbidden: not usable by user or serviceaccount, provider restricted-v2: .spec.securityContext.fsGroup: Invalid value: [
]int64{1001}: 1001 is not an allowed group, provider restricted-v2: .containers[0].runAsUser: Invalid value: 1001: must be in the ranges: [1001010000, 1001
019999], provider "restricted": Forbidden: not usable by user or serviceaccount, provider "nonroot-v2": Forbidden: not usable by user or serviceaccount, pr
ovider "nonroot": Forbidden: not usable by user or serviceaccount, provider "pcap-dedicated-admins": Forbidden: not usable by user or serviceaccount, provi
der "hostmount-anyuid": Forbidden: not usable by user or serviceaccount, provider "machine-api-termination-handler": Forbidden: not usable by user or servi
ceaccount, provider "hostnetwork-v2": Forbidden: not usable by user or serviceaccount, provider "hostnetwork": Forbidden: not usable by user or serviceacco
unt, provider "hostaccess": Forbidden: not usable by user or serviceaccount, provider "splunkforwarder": Forbidden: not usable by user or serviceaccount, p
rovider "trident-node-linux": Forbidden: not usable by user or serviceaccount, provider "node-exporter": Forbidden: not usable by user or serviceaccount, p
rovider "privileged": Forbidden: not usable by user or serviceaccount]
[ec2-user@ip-10-49-11-132 astra]$
```

Fix the error by editing the `runAsUser` and `fsGroup` fields in `statefuleset.apps/postgresql` object with the uid that is in the output of the `oc get project` command as shown below.

```
[ec2-user@ip-10-49-11-132 astra]$ oc get project postgresql -o yaml | grep uid-range
    openshift.io/sa.scc.uid-range: 1001010000/10000
[ec2-user@ip-10-49-11-132 astra]$ oc edit -n postgresql statefulset.apps/postgresql
statefulset.apps/postgresql edited
[ec2-user@ip-10-49-11-132 astra]$
```

postgresql app should be running and using persistent volumes backed by Amazon FSx for NetApp ONTAP storage.

```
[ec2-user@ip-10-49-11-132 astra]$ oc get pods -n postgresql
NAME            READY   STATUS    RESTARTS   AGE
postgresql-0    1/1     Running   0          2m46s
[ec2-user@ip-10-49-11-132 astra]$
```

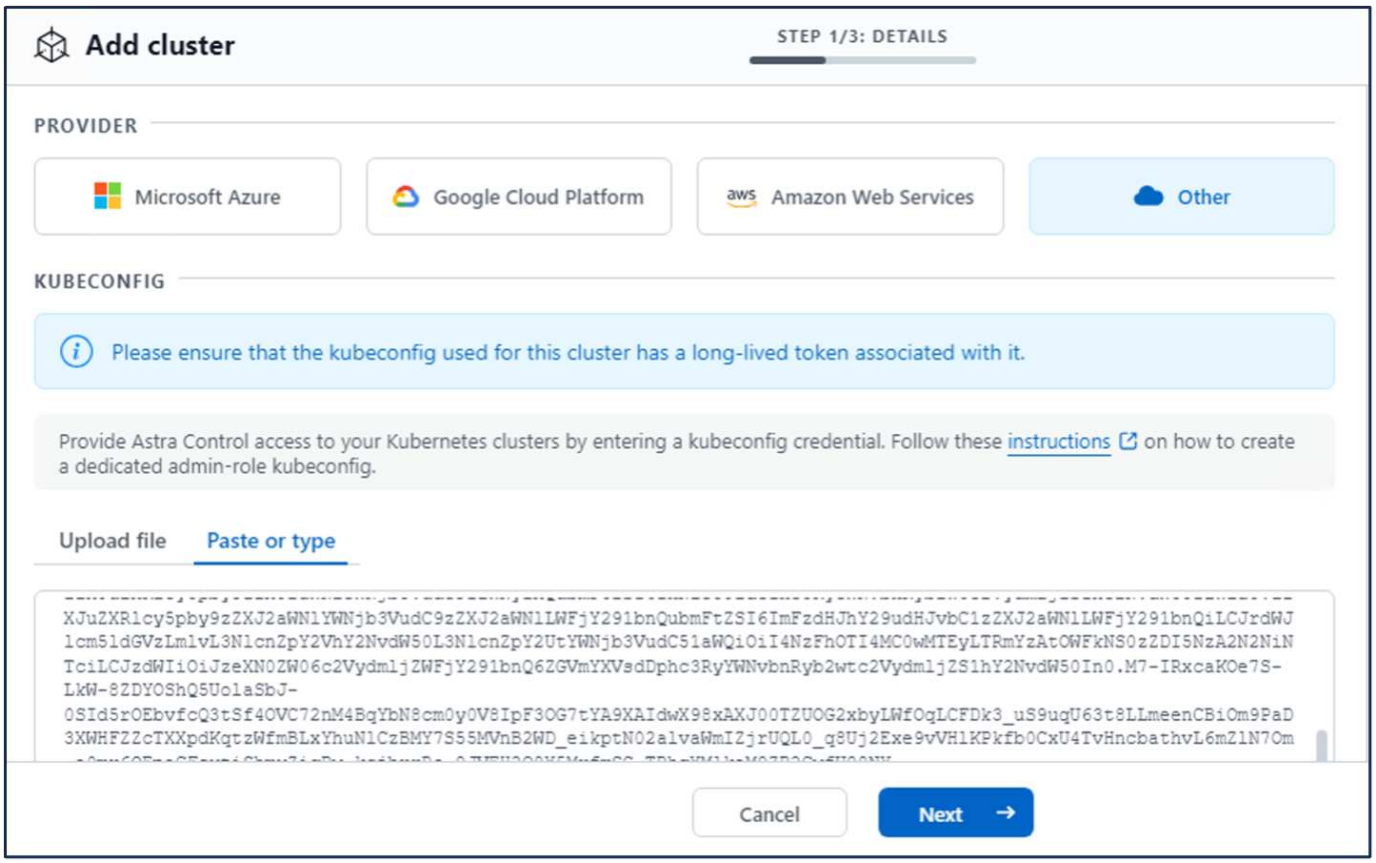
```
[ec2-user@ip-10-49-11-132 storage]$ kubectl get pvc -n postgresql
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS   AGE
data-postgresql-0   Bound    pvc-dd09524a-de75-4825-9424-03a9b91195ca   8Gi        RWO            ontap-nas      4m2s
[ec2-user@ip-10-49-11-132 storage]$
```

6. Create a database and add a record

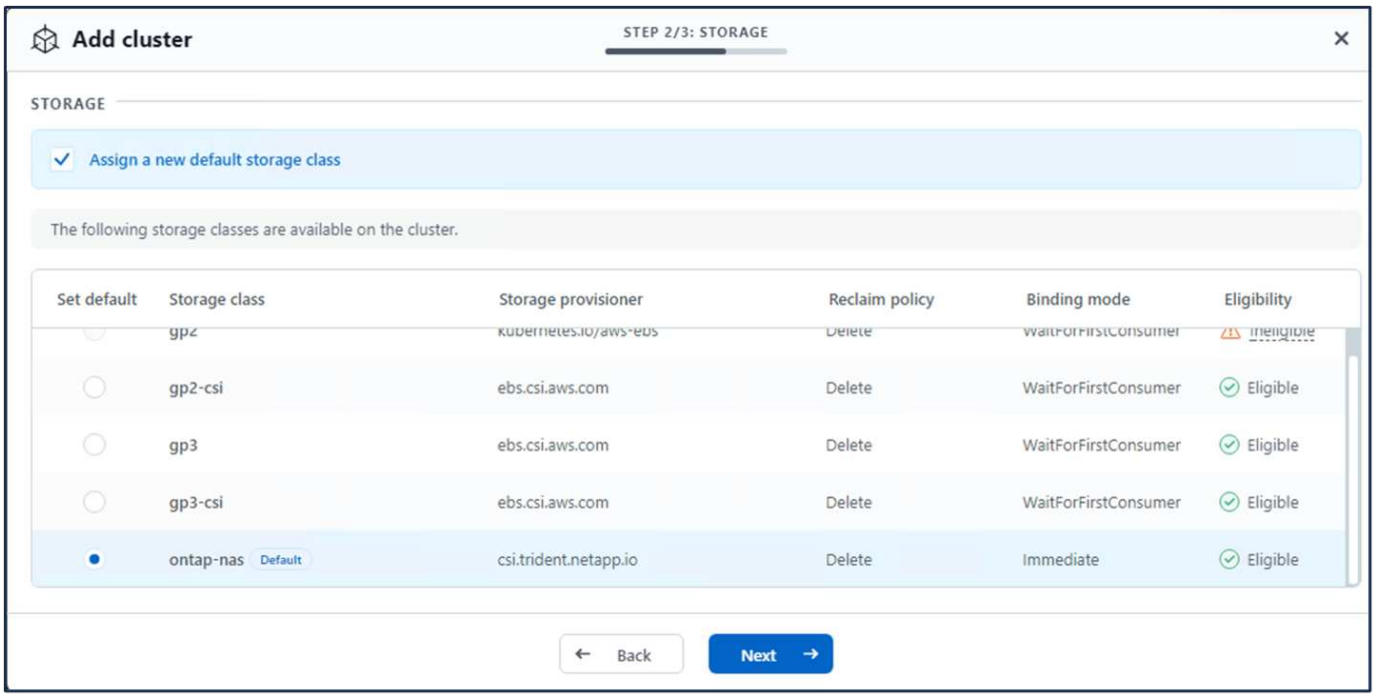
```
[ec2-user@ip-10-49-11-132 astra]$ export POSTGRES_PASSWORD=$(kubectl get secret --namespace postgresql postgresql -o jsonpath="
{.data.postgres-password}" | base64 -d)
[ec2-user@ip-10-49-11-132 astra]$ kubectl run postgresql-client --rm --tty -i --restart='Never' --namespace postgresql --image
docker.io/bitnami/postgresql:16.2.0-debian-11-r1 --env="PGPASSWORD=$POSTGRES_PASSWORD" \
> --command -- psql --host postgresql -U postgres -d postgres -p 5432
Warning: would violate PodSecurity "restricted:v1.24": allowPrivilegeEscalation != false (container "postgresql-client" must se
t securityContext.allowPrivilegeEscalation=false), unrestricted capabilities (container "postgresql-client" must set securityCo
ntext.capabilities.drop=["ALL"]), runAsNonRoot != true (pod or container "postgresql-client" must set securityContext.runAsNonR
oot=true), seccompProfile (pod or container "postgresql-client" must set securityContext.seccompProfile.type to "RuntimeDefault
" or "Localhost")
If you don't see a command prompt, try pressing enter.

postgres=# CREATE DATABASE erp;
CREATE DATABASE
postgres=# \c erp
You are now connected to database "erp" as user "postgres".
erp=# CREATE TABLE PERSONS(ID INT PRIMARY KEY NOT NULL, FIRSTNAME TEXT NOT NULL, LASTNAME TEXT NOT NULL);
CREATE TABLE
erp=# INSERT INTO PERSONS VALUES(1,'John','Doe');
INSERT 0 1
erp=# \dt
          List of relations
 Schema |  Name   | Type  |  Owner
--------+---------+-------+----------
 public | persons | table | postgres
(1 row)

erp=# SELECT * FROM persons;
 id | firstname | lastname
----+-----------+----------
  1 | John      | Doe
(1 row)
```

7. Add the cluster into ACS

Log in to ACS. Select cluster and click on Add. Select other and upload or paste the kubeconfig file.



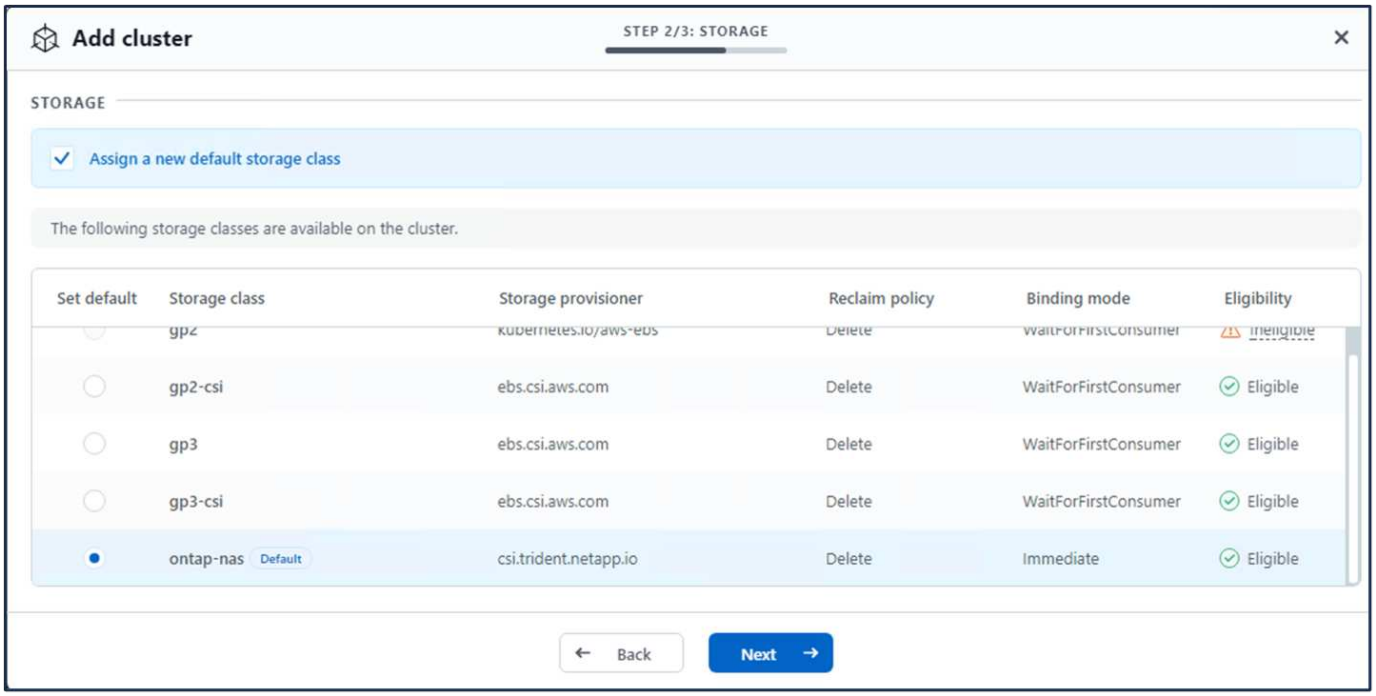
Click **Next** and select ontap-nas as the default storage class for ACS. Click **Next**, review the details and **Add** the cluster.



8. Define the application in ACS

Define the postgresql application in ACS. From the landing page, select **Applications**, **Define** and fill in the appropriate details. Click **Next** a couple of times, Review the details and click **Define**. The application gets

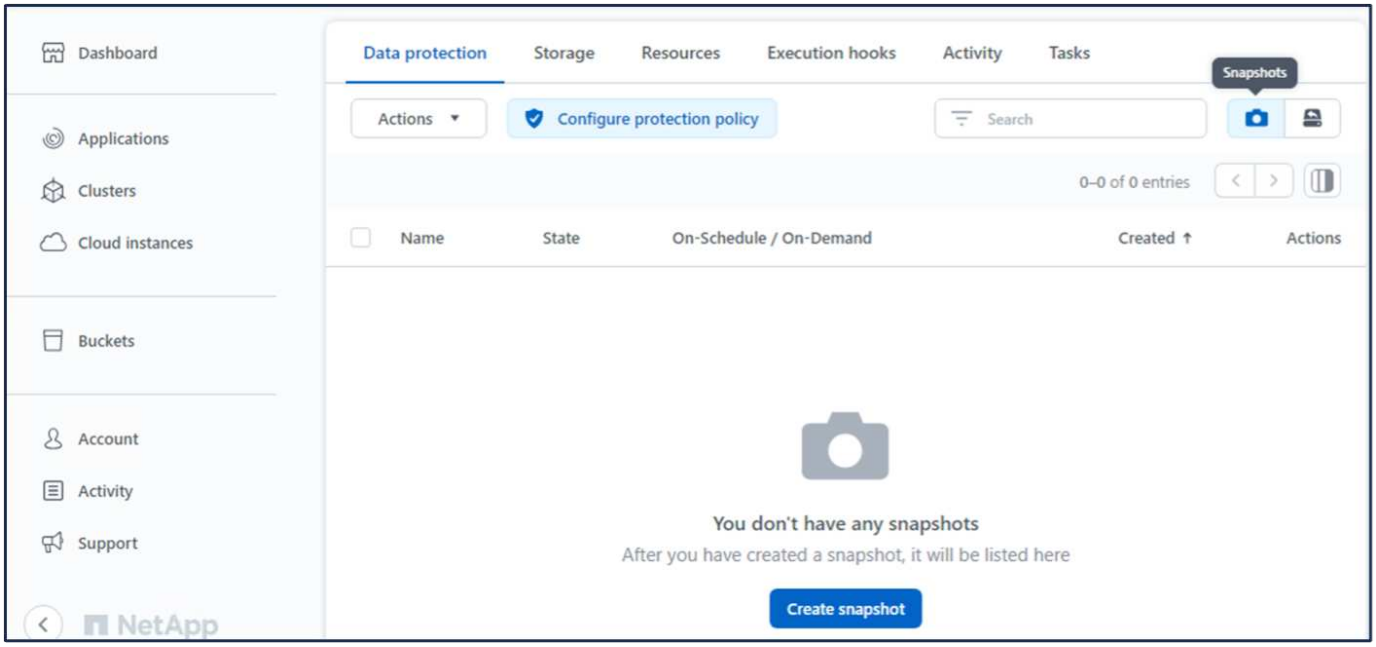
added to ACS.

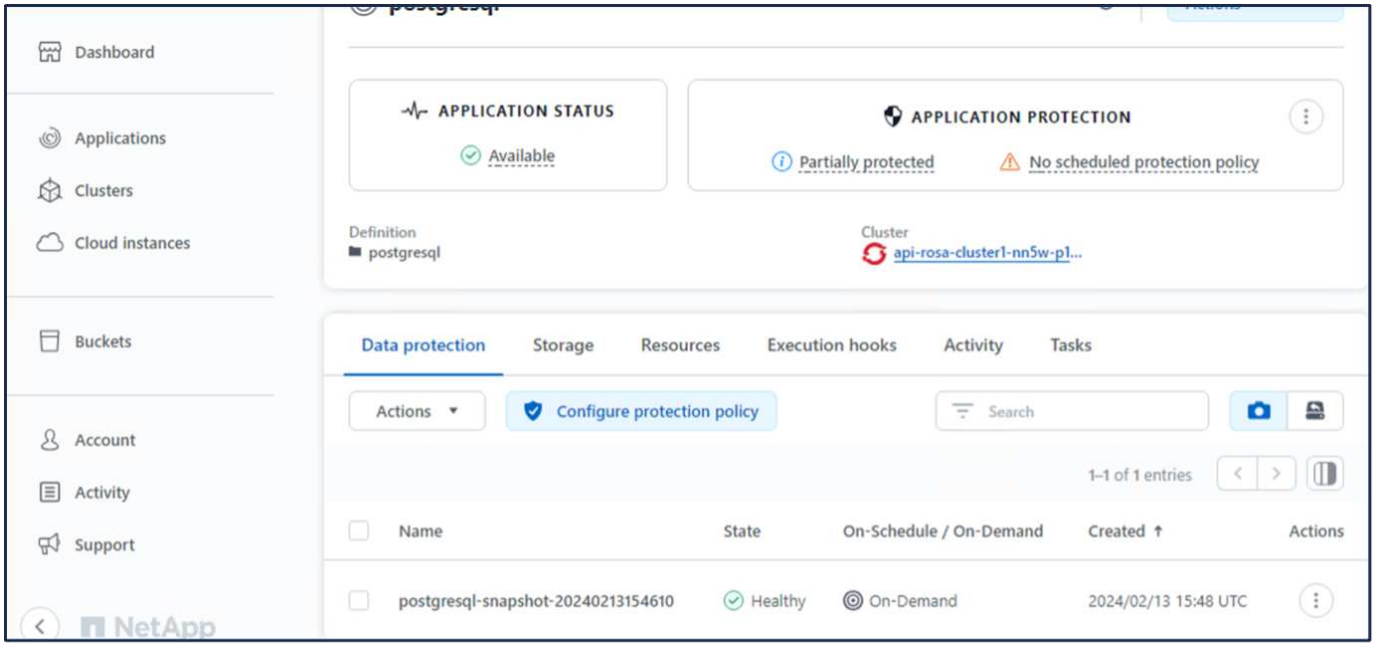


9. Create a snapshot using ACS

There are many ways to create a snapshot in ACS. You can select the application and create a snapshot from the page that shows the details of the application. You can click on Create snapshot to create an on-demand snapshot or configure a protection policy.

Create an on-demand snapshot by simply clicking on **Create snapshot**, providing a name, reviewing the details, and clicking on **Snapshot**. The snapshot state changes to Healthy after the operation is completed.





10. Delete the database in the postgresql application

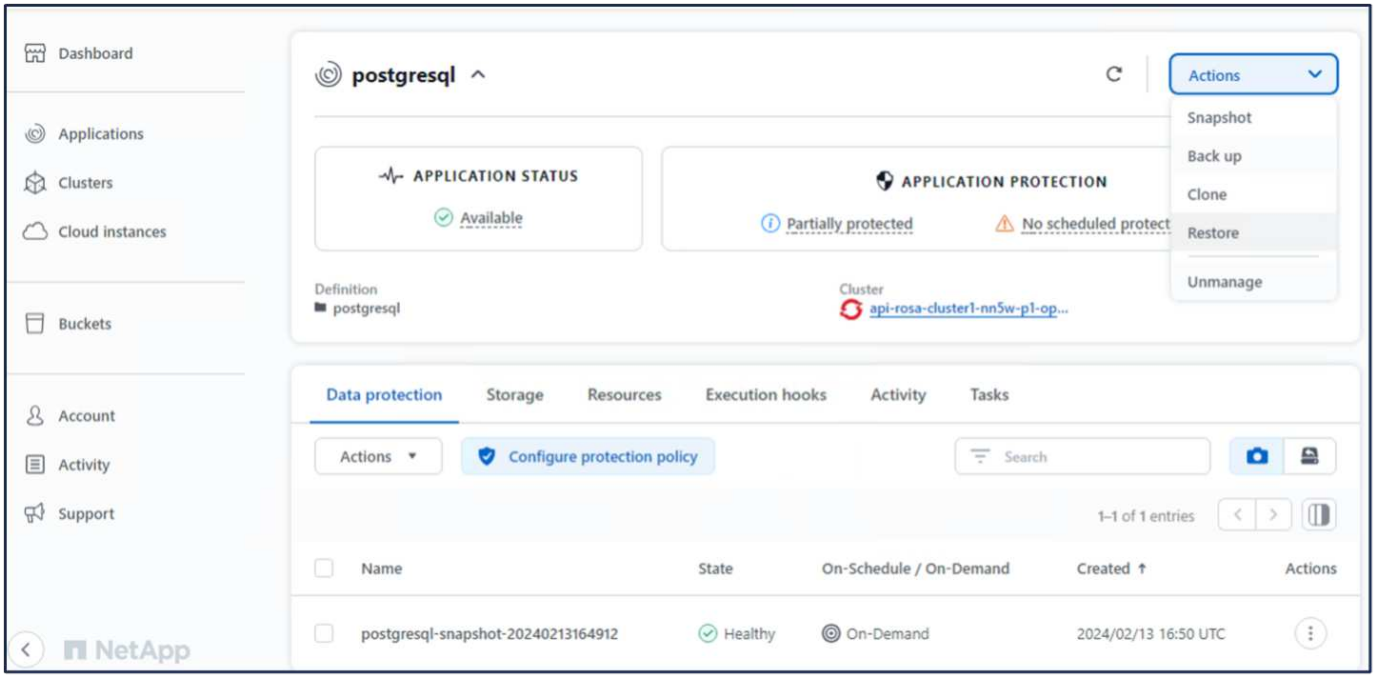
Log back into postgresql, list the available databases, delete the one you created previously and list again to ensure that the database has been deleted.

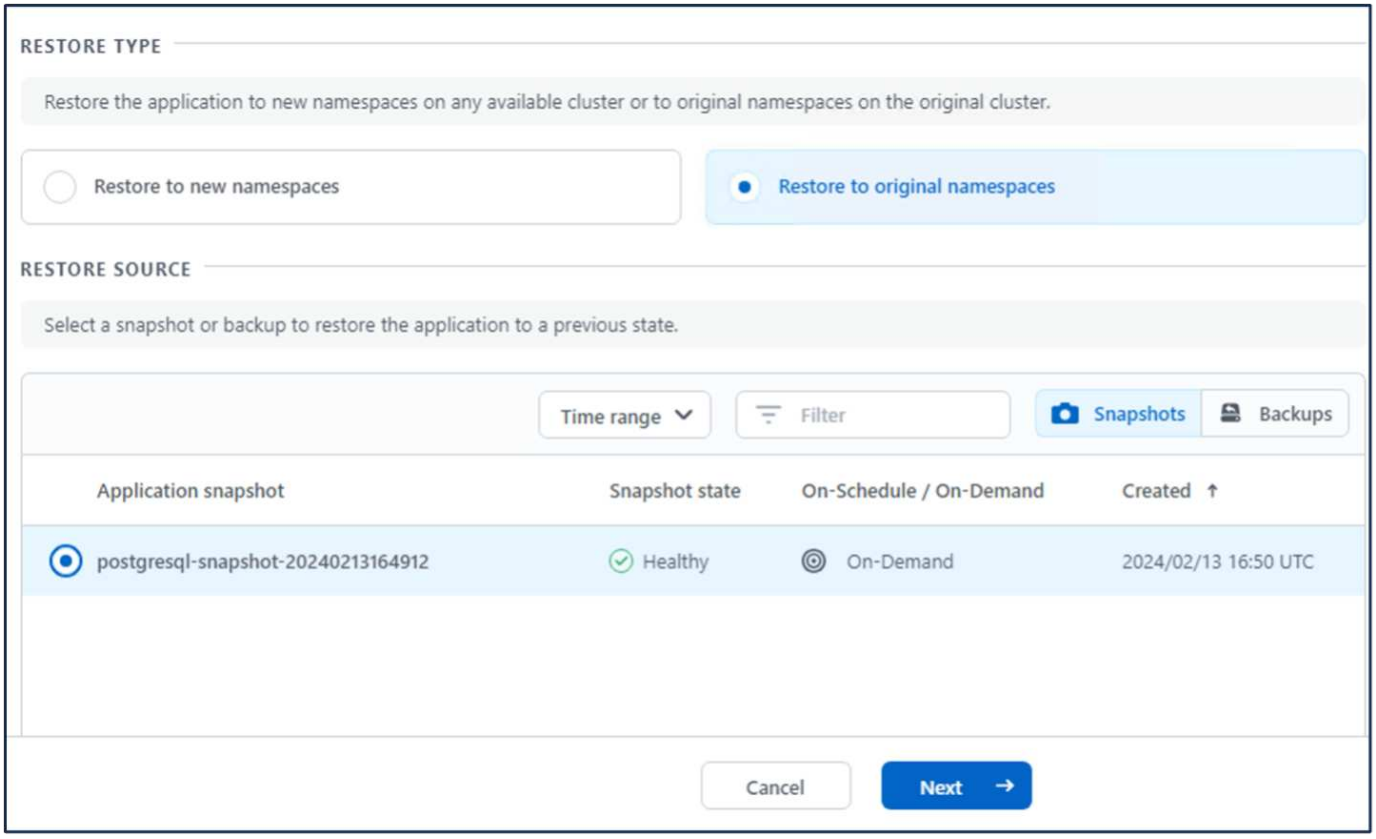
```
postgres=# \l
                                                    List of databases
   Name    |  Owner   | Encoding | Locale Provider |   Collate   |    Ctype    | ICU Locale | ICU Rules |   Access priv
-----------+----------+----------+-----------------+-------------+-------------+------------+-----------+---------------
-
 erp       | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           |
 postgres  | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           |
 template0 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres
+
           |          |          |                 |             |             |            |           | postgres=CTc/
 template1 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres
+
           |          |          |                 |             |             |            |           | postgres=CTc/
(4 rows)

postgres=# DROP DATABASE erp;
DROP DATABASE
postgres=# \l
                                                    List of databases
   Name    |  Owner   | Encoding | Locale Provider |   Collate   |    Ctype    | ICU Locale | ICU Rules |   Access priv
-----------+----------+----------+-----------------+-------------+-------------+------------+-----------+---------------
-
 postgres  | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           |
 template0 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres
+
           |          |          |                 |             |             |            |           | postgres=CTc/
 template1 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres
+
           |          |          |                 |             |             |            |           | postgres=CTc/
(3 rows)
```

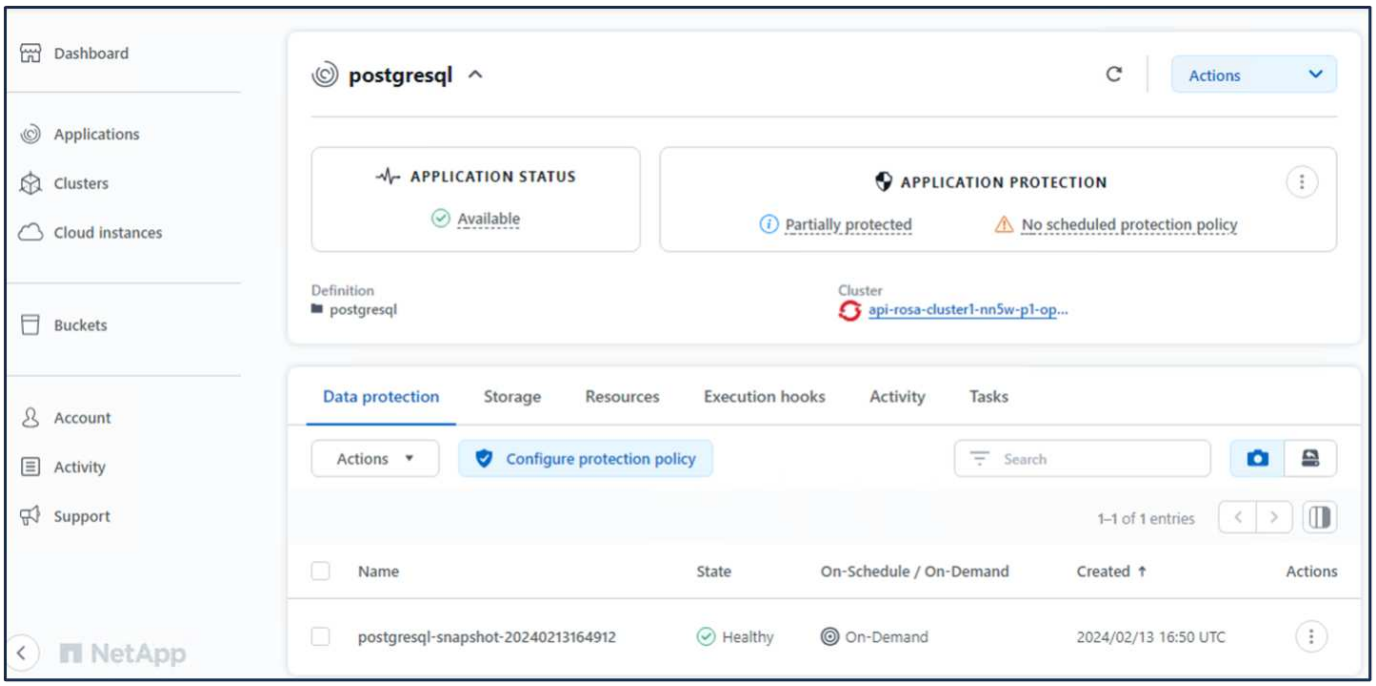
11. Restore from a snapshot using ACS

To restore the application from a snapshot, go to ACS UI landing page, select the application and select Restore. You need to pick a snapshot or a backup from which to restore. (Typically, you would have multiple

created based on a policy that you have configured). Make appropriate choices in the next couple of screens and then click on **Restore**. The application status moves from Restoring to Available after it has been restored from the snapshot.







12. Verify your app has been restored from the snapshot

Login to the postgresql client and you should now see the table and the record in the table that you previously had. That's it. Just by clicking a button, your application has been restored to a previous state. That is how easy we make it for our customers with Astra Control.

```
[ec2-user@ip-10-49-11-132 ~]$ kubectl run postgresql-client --rm --tty -i --restart='Never' --namespace postgresql --image docker.io/bitnami/postgresql:16.
2.0-debian-11-r1 --env="PGPASSWORD=$POSTGRES_PASSWORD" --command -- psql --host postgresql -U postgres -d postgres -p 5432
Warning: would violate PodSecurity "restricted:v1.24": allowPrivilegeEscalation != false (container "postgresql-client" must set securityContext.allowPrivi
legeEscalation=false), unrestricted capabilities (container "postgresql-client" must set securityContext.capabilities.drop=["ALL"]), runAsNonRoot != true (
pod or container "postgresql-client" must set securityContext.runAsNonRoot=true), seccompProfile (pod or container "postgresql-client" must set securityCon
text.seccompProfile.type to "RuntimeDefault" or "Localhost")
If you don't see a command prompt, try pressing enter.

postgres=# \l
                                                   List of databases
   Name    |  Owner   | Encoding | Locale Provider |   Collate   |    Ctype    | ICU Locale | ICU Rules |   Access privileges
-----------+----------+----------+-----------------+-------------+-------------+------------+-----------+-----------------------
 erp       | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           |
 postgres  | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           |
 template0 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres          +
           |          |          |                 |             |             |            |           | postgres=CTc/postgres
 template1 | postgres | UTF8     | libc            | en_US.UTF-8 | en_US.UTF-8 |            |           | =c/postgres          +
           |          |          |                 |             |             |            |           | postgres=CTc/postgres
(4 rows)

postgres=# \c erp
You are now connected to database "erp" as user "postgres".
erp=# \dt
          List of relations
 Schema |  Name   | Type  |  Owner
--------+---------+-------+----------
 public | persons | table | postgres
(1 row)

erp=# SELECT * from PERSONS;
 id | firstname | lastname
----+-----------+----------
  1 | John      | Doe
(1 row)
```

Data migration

This page shows the data migration options for container workloads on Managed Red Hat OpenShift clusters using FSx for NetApp ONTAP for persistent storage.

Data Migration

Red Hat OpenShift service on AWS as well as FSx for NetApp ONTAP (FSxN) are part of their service portfolio by AWS. FSxN is available on Single AZ or Multi-AZ options.
Multi-Az option provides data protection from availability zone failure.
FSxN can be integrated with Astra Trident to provide persistent storage for applications on ROSA clusters.

Integration of FSxN with Trident using Helm chart

ROSA Cluster Integration with Amazon FSx for ONTAP

The migration of container applications involves:

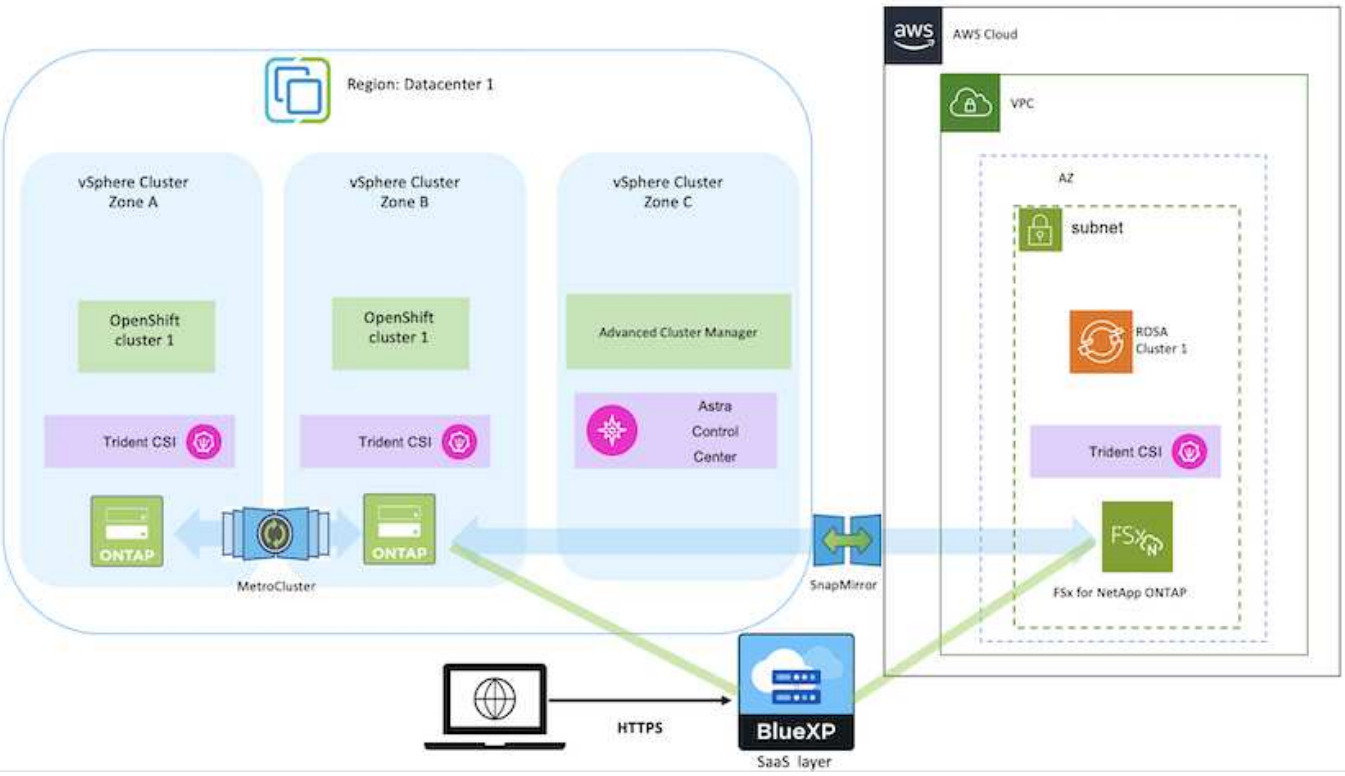
- Persistent volumes: this can be accomplished using BlueXP.
  Another option is to use Astra Control Center to handle container application migrations from on-premises to the cloud environment. Automation can be used for the same purpose.
- Application metadata: this can be accomplished using OpenShift GitOps (Argo CD).

Failover and Fail-back of applications on ROSA cluster using FSxN for persistent storage

The following video is a demonstration of application failover and fail-back scenarios using BlueXP and Argo CD.

Failover and Fail-back of applications on ROSA cluster

Data protection and migration solution for OpenShift Container workloads



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